

why ten?

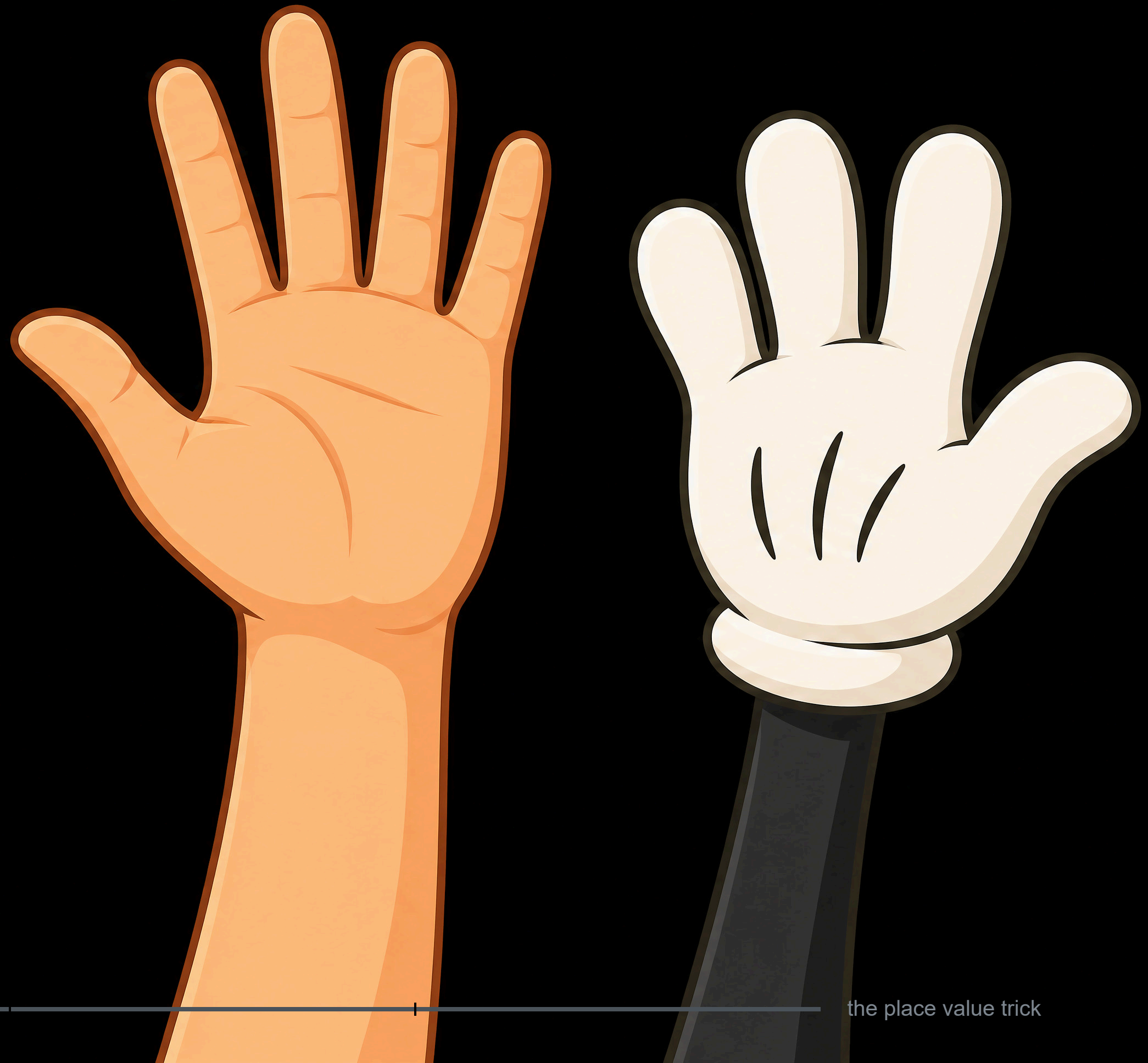
digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 the place value trick
- 2 why two?
- 3 reading bytes
- 4 your own base

part 1

the place value trick

why do we
count to ten?



what 123 really says

base 10: powers of ten

1

×

100

2

×

10

3

×

1

what 123 really says

base 10: powers of ten



100

10

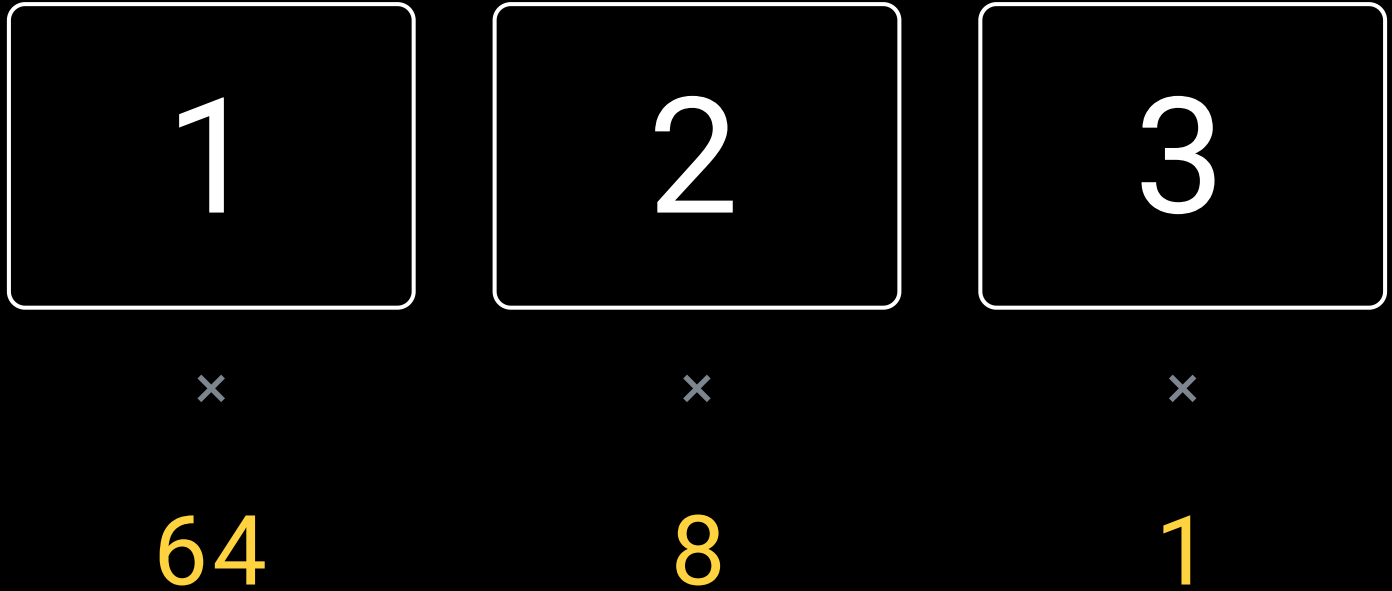
1

$$1 \cdot 100 + 2 \cdot 10 + 3 \cdot 1 = 123$$

every place carries a power of ten.

the eight-finger creature writes 123

base 8: powers of eight



the eight-finger creature writes 123

base 8: powers of eight



×

×

×

64

8

1

$$1 \cdot 64 + 2 \cdot 8 + 3 \cdot 1 = 83$$

same recipe. different base. different number.

and a creature
with two
flippers?



two flippers: counting in binary

we write	the dolphin writes
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000

a new place opens
when the digits run out,
just like 9 to 10

the dolphin writes 110

base 2: powers of two



the dolphin writes 110

base 2: powers of two



×

×

×

4

2

1

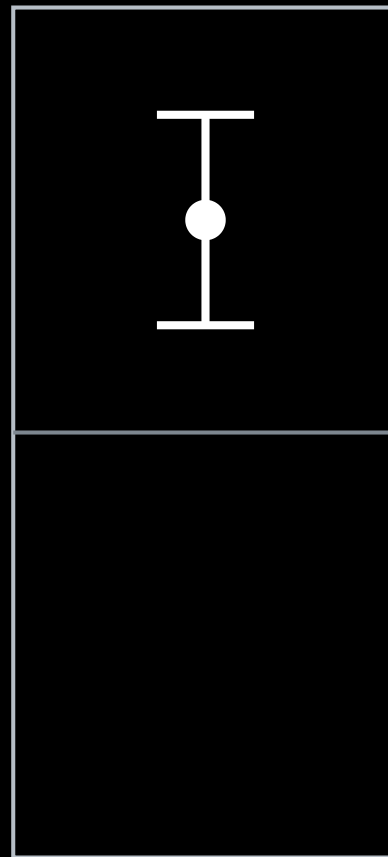
$$1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 = 6$$

nothing new. the third time is the same trick.

part 2

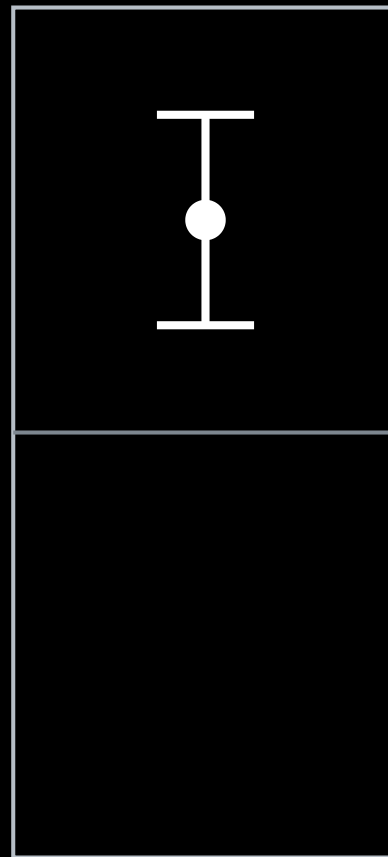
why two?

why two, of all things?



2 states

why two, of all things?



2 states



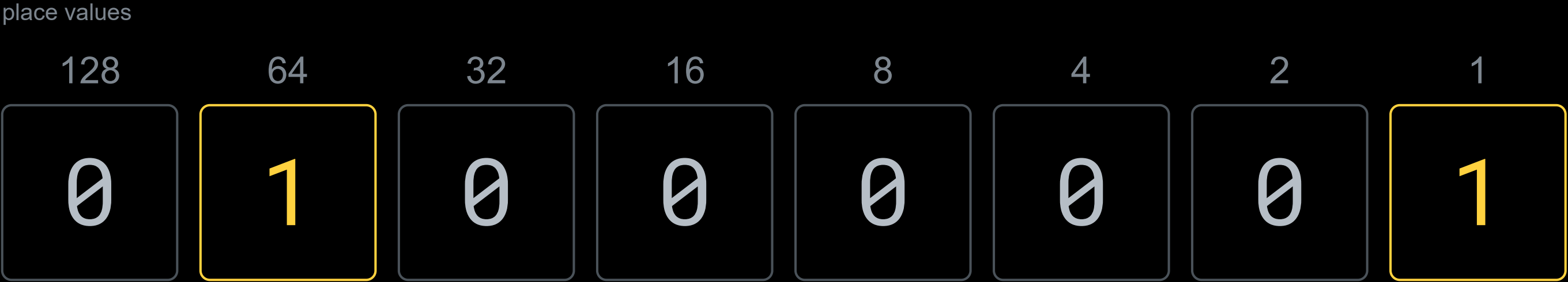
10 states

the same wobble on the wire.
with two states it never matters,
with ten it always does.

the price: more places.
places are cheap.

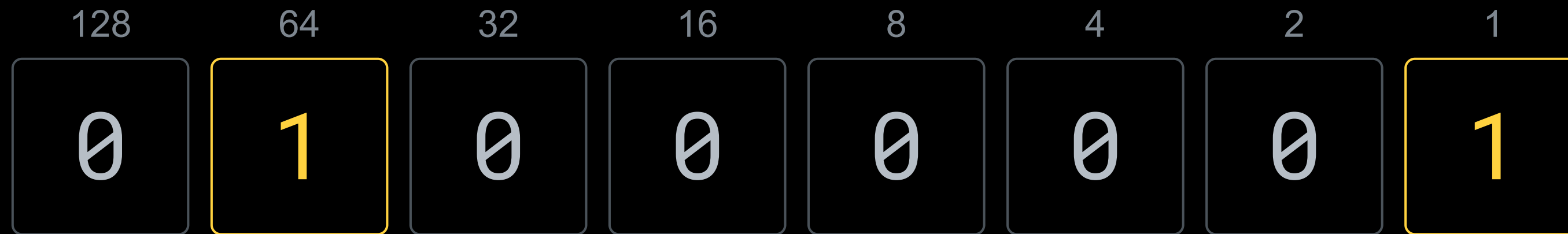
two states are the cheapest thing you can build reliably.

eight bits make a byte



eight bits make a byte

place values



$$64 + 1 = 65$$

$2^8 = 256$ values, 0 to 255

256 values, and every one of them is a place value sum.

try it: eight switches, one number

2⁷
128

0

2⁶
64
1

2⁵
32

0

2⁴
16

0

2³
8

0

2²
4

0

2¹
2

0

2⁰
1
1

= 64 + 1 = 65

binary

0100
0001
grouped in two nibbles of four bits

-1

+1

random

reset

decimal

65
0 to 255

hex

41
one digit per nibble

as text (ascii)

A
character number 65 in the ascii agreement

why two?

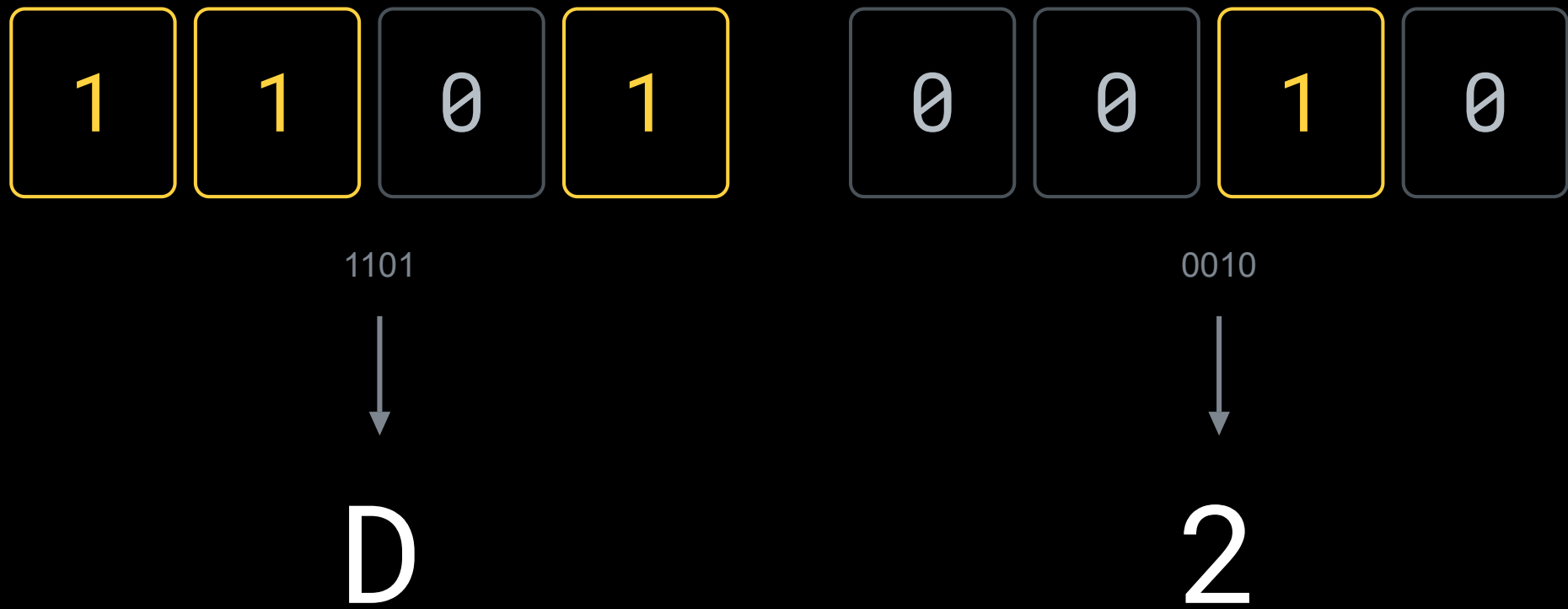
part 3

reading bytes

hex: four bits at a time

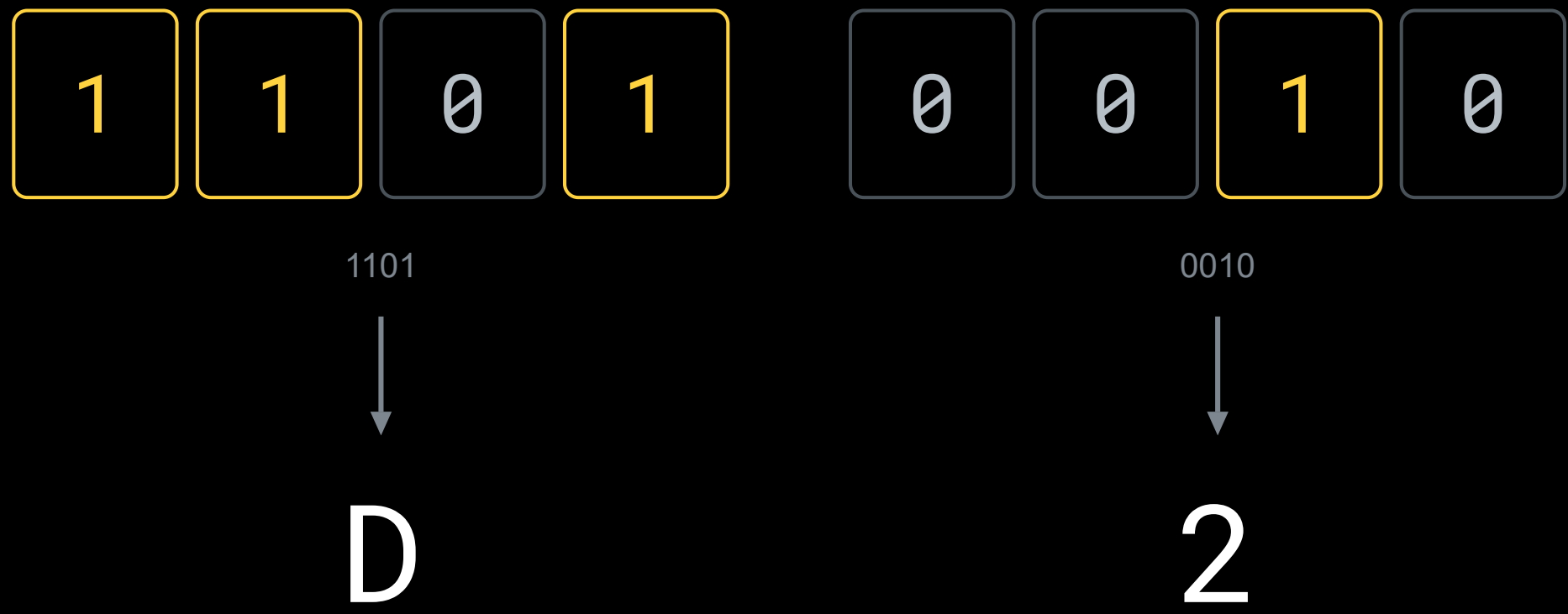


hex: four bits at a time



four bits have 16 states.
so one hex digit fits a nibble,
and a byte fits two.

hex: four bits at a time



four bits have 16 states.
so one hex digit fits a nibble,
and a byte fits two.

ten digits are not enough, so six letters help out

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

hex is not a new number world. it is a reading aid.

try it: the same file, twice

picture:   

pixels:

8 × 8

16 × 16

32 × 32

what a viewer makes of the bytes



click a pixel

open as text

reset file

the bytes (a hex editor)

0000	42	4d	36	03	00	00	00	00	00	00	36	00	00	00	28	00
0010	00	00	10	00	00	00	10	00	00	00	01	00	18	00	00	00
0020	00	00	00	03	00	00	c4	0e	00	00	c4	0e	00	00	00	00
0030	00	00	00	00	00	00	0a	0b	17	0e	11	1c	6b	69	62	45
0040	41	41	11	16	1e	12	19	20	13	19	20	10	18	1e	10	1a
0050	44	1b	22	42	0d	17	82	19	21	b0	1b	25	e4	26	3a	f6
0060	28	3d	f9	23	36	f9	0c	0d	1a	56	5e	66	a2	a4	a1	38
0070	3a	3e	10	17	1e	12	1c	21	1d	27	31	11	13	21	0b	0b
0080	1d	06	06	18	2e	33	62	28	30	91	1b	1e	da	1e	26	e6
0090	26	3a	f2	27	37	f5	21	25	32	9e	a8	ae	a2	a9	ab	55
00a0	57	55	16	17	1a	31	2e	31	3a	33	36	2a	24	27	25	20
00b0	23	1b	16	1f	40	3a	45	a2	94	a8	4b	4d	c6	18	12	e0
00c0	17	0c	df	12	0b	da	50	57	62	9e	ab	b6	a1	a9	af	6a
00d0	69	68	12	10	11	4d	48	49	2d	27	26	2e	27	27	24	1e
00e0	20	25	1f	23	96	89	8e	c5	b5	bf	97	90	c5	22	29	ee
00f0	1d	18	eb	1a	13	e9	83	8f	9c	9d	ad	bb	a2	ad	b5	90
0100	8e	88	42	3a	39	4d	45	43	2c	26	25	2c	25	25	2d	26
0110	26	75	6b	6e	ca	bb	be	ce	bf	c5	ca	bd	c7	72	7f	de
0120	1f	28	f2	1a	24	f2	00	c7	b0	0e	eb	bd	08	e0	b7	0e

hover a byte

- ☐ file header
- ☐ info header
- ☐ pixel data
- ☒ selected pixel

822 bytes = 14 file header + 40 info header + 768 pixel data (16 × 16 pixels × 3 bytes)

photos: ai-generated (gpt-image-2)

two ladders of units

decimal: each step × 1000

binary: each step × 1024

byte	1	1
kilobyte	1 000	1 024
megabyte	1 000 000	1 048 576
gigabyte	1 000 000 000	1 073 741 824
terabyte	1 000 000 000 000	1 099 511 627 776

both are called the same in everyday speech. that is where the trouble starts.

the missing 69 gigabytes

on the box:
1 TB

1 000 000 000 000 bytes

counted up the decimal ladder, × 1000

the missing 69 gigabytes



nobody cheated. two ladders, one word.

part 4

your own base

you built one already

the agreed digits

0

1

2

3

one letter, three colour positions

×

4^2

×

4^1

×

4^0

you built one already

the agreed digits



one letter, three colour positions



×

4^2



×

4^1



×

4^0

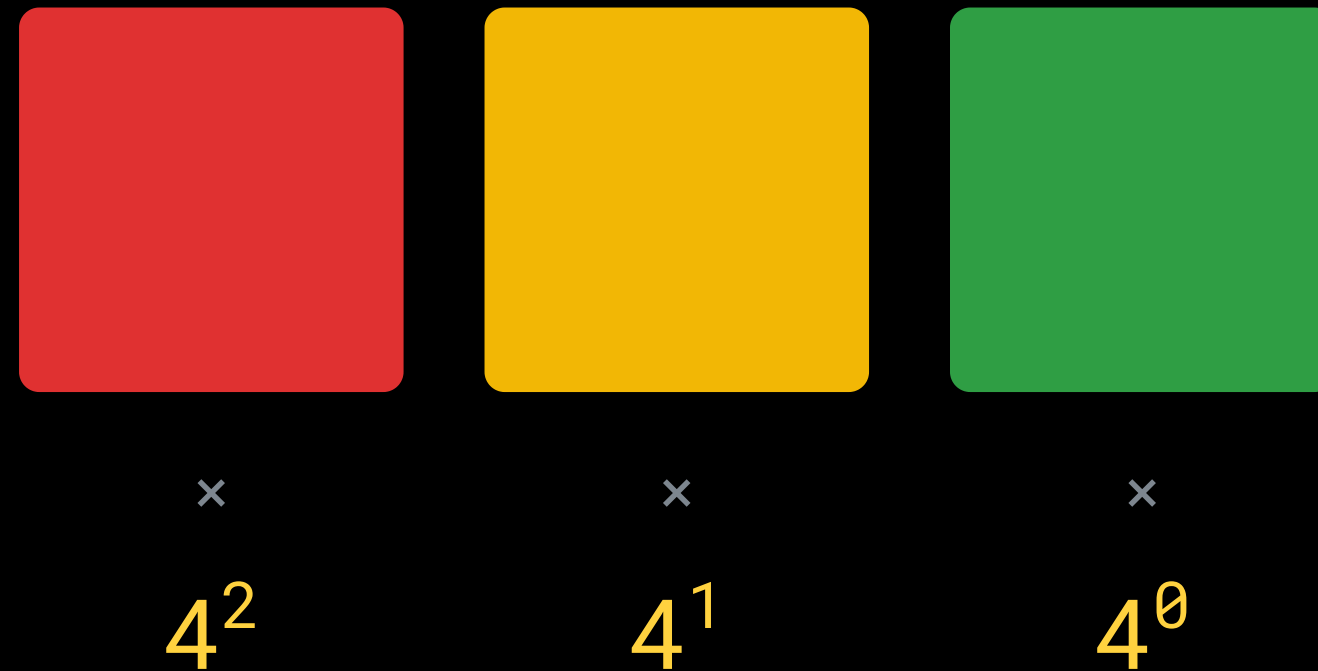
$$0 \cdot 16 + 3 \cdot 4 + 1 \cdot 1 = 13$$

you built one already

the agreed digits



one letter, three colour positions



$$0 \cdot 16 + 3 \cdot 4 + 1 \cdot 1 = 13$$

$4^3 = 64$ combinations, enough for an alphabet

k colours are k digits, and position matters.

ten was never special.

ten was never special.

it is the number of fingers on two hands, and nothing else.